

# THE WANING BORDER

*Version 1.2*

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## **Game Design Document**

Unity 6 DOTS/ECS Real-Time Strategy Game

March 2026

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# 1. GAME OVERVIEW

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**The Waning Border** is a real-time strategy game built on Unity 6 using the Data-Oriented Technology Stack (DOTS) and Entity Component System (ECS) architecture. Set in a dark fantasy world where ancient civilizations clash for dominance while an encroaching crystalline corruption threatens to consume everything in its path.

Players command factions, gather resources across four tiers (Supplies, Iron, Crystal, Veilsteel), build fortified bases, research technologies through branching tech trees, train diverse armies, and battle for survival against both rival factions and the ever-expanding Crystal threat.

The game features three distinct playable cultures - **Runai** (trade-focused mercantile power), **Alanthor** (defensive fortress builders with heavy units), and **Feraldis** (aggressive war economy fueled by pillage) - each with unique military rosters, economic mechanics, and strategic identity.

Beyond cultures, players adopt one of **12+ religious sects**, each unlocking a unique specialist unit trained from the Temple. Combined with a full AI opponent system supporting multiple personalities and difficulty levels, fog of war, battalion formations with synchronized movement, and a unified command routing system, The Waning Border delivers deep strategic gameplay with modern technical foundations.

|                  |                                                 |
|------------------|-------------------------------------------------|
| Engine           | Unity 6 (6000.0.37f1)                           |
| Architecture     | DOTS / ECS (Entities 1.3.14)                    |
| Genre            | Real-Time Strategy                              |
| Cultures         | Runai, Alanthor, Feraldis                       |
| Sects            | 12+ Religious Orders                            |
| Unit Types       | 35 unique units                                 |
| Resources        | Supplies, Iron, Crystal, Veilsteel              |
| AI Difficulties  | Easy, Normal, Hard, Expert                      |
| AI Personalities | Balanced, Aggressive, Defensive, Economic, Rush |

## 2. FEATURES

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19 major feature systems. 18 fully implemented and stable; 1 (Multiplayer) has its foundation in place with the networking layer pending.

# Resource Economy

[COMPLETE]

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Four-tier resource system forming the economic backbone of The Waning Border. **Supplies** are generated passively from population. **Iron** and **Crystal** are gathered by Miner units from map deposits - miners travel to deposits, extract resources over time, and carry them back to the Hall or Gatherer Hut. When a deposit is depleted, miners automatically search for new deposits within their line-of-sight range. **Veilsteel** serves as an advanced resource for late-game technologies.

- Iron mining: 1 iron per 2 seconds, max carry capacity of 10
- Crystal mining: 1 crystal per 1.5 seconds, max carry capacity of 10
- Supplies income scales with population count
- Local player miners require explicit Gather commands; AI miners auto-find deposits
- Vault buildings generate passive interest income (Alanthor)

# Base Building & Construction

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Players construct buildings using Builder units on a passability grid with full terrain validation. Multiple builders can work simultaneously (each contributes 1.0 progress per second). Buildings cannot be placed on slopes, water, or overlapping existing structures.

- Builders auto-chain to nearby unfinished structures within line-of-sight
- Shift+click enables repeated placement mode for rapid base expansion
- Buildings: Hall, Barracks, Gatherer Hut, Hut, Smelter, Trade Hub, Outpost, Temple
- Building repair system restores damaged structures
- Passability grid updates dynamically when buildings are placed or destroyed
- Deferred defense system activates building combat after construction completes

# Unit Training & Population

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Units are trained from production buildings with a queue-based system. The Hall produces economy units (Builders, Miners, Scouts), while the Barracks trains military units. Each unit has specific training costs and time. Population is tracked against capacity - Huts provide housing slots.

- Feraldis batch training: 5 units simultaneously with 10% time discount, 5% cost discount
- Berserker conversion transforms Miners into combat units at the Fiendstone Keep
- Rally points set where newly trained units gather after spawning
- Training queue supports multiple items with automatic progression
- Population sync system prevents over-spawning in edge cases

# Combat System

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Comprehensive combat with a damage-type vs armor-type modifier matrix. Four damage types (Melee, Ranged/Pierce, Siege, Magic) interact with per-type defense using diminishing returns. Armor types (InfantryLight, InfantryHeavy, Cavalry, Ranged, Structure) determine base modifiers. Height advantage grants up to +/-20% damage.

- Melee combat: chase behavior, minimum 1 damage guarantee, crystal buff/debuff integration
- Ranged combat: aim timer, LoS validation, auto-retreat when enemies close in (ArcherTag)
- Projectiles: arrows use quadratic Bezier curves, lasers use straight-line flight
- Spell system: timed buffs (armor, damage multiplier, speed) and debuffs (slow, drain)
- Invulnerability mechanic with timed expiry
- Building combat: turrets and walls with defensive fire
- Litharch healing: 8 HP/sec to wounded allies within range 4



# Movement & Pathfinding

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Flow-field based pathfinding with LRU cache (8 fields). Destination snapping for deduplication and grid version tracking for staleness detection. Multi-frame BFS generation prevents frame spikes.

- Terrain height snapping keeps units grounded on slopes
- Slope and water blockage prevents invalid movement
- 3-tier stuck detection escalation resolves pathfinding deadlocks
- Smooth direction lerping prevents oscillation during movement
- Unit separation system applies repulsion forces to prevent clustering
- Patrol system: ping-pong waypoint cycling with auto-engagement

# Battalion Formations

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Units form battalions (5 columns x 3 rows, 1.5m spacing) led by a Battalion Leader. Members synchronize movement using ratio-based speed scaling so all arrive simultaneously. Two-pass algorithm: compute slots + max distance, then scale each member's speed.

- Leash system constrains members to maximum separation from leader
- Three stances: Default (balanced), Defensive (hold ground), Aggressive (pursue)
- Battalion leaders excluded from targeting system to protect formation integrity
- Formation converges all members simultaneously for clean movement

# Cultures System

[COMPLETE]

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Three cultures unlocked via Hall age-up, altering military roster, economy, and identity. Each provides 4-6 unique military units and culture-specific economic bonuses. Culture selection does not change the player's faction color.

- Runai: Trade economy with Caravans, Raiders, Spearmen, Skirmishers, Sand Ballistas
- Alanthor: Defensive powerhouse with Sentinels, Crossbowmen, Cataphracts, Ballistas
- Feraldis: War economy with Warboar Riders, Hunters, Siege Rams, Berserkers
- Hall age-up: ticks AgeUpState, sets culture, scales hall 1.3x, advances to Era 2
- Alanthor Gatherer Huts have self-destruct timers after age-up

# Runai Trade Economy

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Exclusive Runai economic system built around automated trade routes. Routes discovered every 2 seconds between same-faction Trade Hubs and nearest Outpost. Caravans spawn automatically (max 3 per route, every 22 seconds) generating income based on route length.

- Income formula:  $25 \times (\text{routeLength} / 30.0)$  supplies per trip
- +25% tariff bonus for routes exceeding 60 distance units
- Caravans receive auto-assigned escorts for protection
- Killing a caravan drops 50% of carried cargo
- Caravans are uncontrollable - fully autonomous
- Caravan escorts die when their paired caravan is destroyed

# Feraldis Pillage Economy

[COMPLETE]

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Feraldis battle economy rewards aggressive military action with direct resource income from combat kills. Every enemy killed or building destroyed grants immediate resources.

- Non-military unit kill: +15 Supplies, +1 Iron
- Military unit kill: +5 Supplies
- Building destruction: +50 Supplies, +5 Iron
- Runs before DeathSystem to capture kill credit
- Checks LastDamagedByFaction to attribute kills to Feraldis players

# Technology Research

[COMPLETE]

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Full tech tree with branching prerequisite chains. Technologies researched from the Temple, each with specific costs, times, and dependencies. Completed research applies stat bonuses and unlocks abilities. Temple upgrades (3 levels) and chapel construction provide bonuses.

- Research queue with automatic progression to next tech
- TechTreeDB stores all tech definitions, costs, times, and dependencies
- TechEffectSystem applies bonuses to existing and future units/buildings
- Temple upgrade progression (3 levels) with chapel sub-buildings
- Chapel cascade destruction when parent temple is destroyed
- Age-up grants Research Points if Temple exists

# Sects (Religious Orders)

[COMPLETE]

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12+ religious sects provide deep customization beyond culture selection. Each sect unlocks a unique specialist unit trained from the Temple, offering tactical options from anti-structure siege to glass-cannon mages to nigh-indestructible tanks.

- Ember Ash: Ashblade (fast melee) | Hollow Brand: Brandbreaker (siege melee)
- Still Flame: Flame Warden (balanced) | Flamewrought Chains: Chaincaster (magic)
- Veiled Memory: Archivist Adept (magic) | Mirror Rite: Glassmark Arcanist (glass cannon)
- Antiquity: Golem Autark (magic tank) | Shard Judgment: Judicator (heavy melee)
- Unmaker's Grasp: Nullblade (anti-Crystal) | Renewal: Scar Guard (tough melee)
- Living Stone: Stone Warden (supreme tank) | Quiet Vault: Vault Keeper (guardian)

# Crystal Faction (AI-Controlled Threat)

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Autonomous AI-controlled Crystal faction (Faction White) as a persistent environmental threat. Operates via its own CrystalAISystem with 5-second decision ticks, independent from standard AI. Crystal Main Nodes build sub-nodes, spawn creatures, send waves, and expand.

- Cursed ground spreads procedurally, damaging units standing on it
- Crystal income: 0.1 crystal per cursed area unit per second
- Spawns Crystallings (swarm), Veilstingers (ranged), Godsplinters (siege boss)
- Sub-nodes: Enforcement (combat), Suppression (debuff), Restoration (heal)
- Crystal death drops reward players for destroying crystal creatures
- Expansion via new Crystal Main Nodes in unexplored cursed areas



# AI Opponent System

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Modular AI with five specialized managers and three behavior modules. Supports multiple personalities adjusting priorities, and difficulty levels modifying numerical advantages.

- Managers: Economy, Military, Building, Tactical, Mission
- Behaviors: Scouting (map exploration), Defense (base protection), Crystal Hunt
- Personalities: Balanced, Aggressive, Defensive, Economic, Rush
- Difficulty: Easy/Normal/Hard/Expert (tuning knobs for units, costs, timing)
- AISharedKnowledge tracks enemy positions, strength, known bases
- Commands route through CommandRouter for consistency

# Fog of War

[COMPLETE]

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Per-faction fog of war with three visibility states. Updated every frame based on unit line-of-sight circles. Affects targeting, minimap, and strategic information.

- Hidden: Never explored, completely dark
- Revealed: Previously seen, greyed out (shows terrain/buildings but not units)
- Visible: Currently within line-of-sight, full information
- Affects targeting - cannot acquire targets in non-visible areas
- Minimap integration shows revealed vs visible areas
- Per-frame visibility stamping from all LOS-bearing units

# Command System

[COMPLETE]

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All commands flow through a single unified CommandRouter ensuring consistent behavior regardless of source and enabling multiplayer synchronization via lockstep queueing.

- Commands: Move, Attack, Attack-Move, Build, Gather, Patrol, Hold Position, Heal, Repair, Convert
- Sources: Local Player, AI, Remote Player (multiplayer), System
- Lockstep queueing for deterministic multiplayer synchronization
- Each command type has an ECS component + static helper class pattern
- Right-click context: Hall/GathererHut with miners = force return/deposit

# Input & Controls

[COMPLETE]

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Full RTS input with standard conventions. Mouse-driven selection and commanding with keyboard modifiers. Control groups for rapid army management, smooth camera navigation.

- Left-click: select | Drag: box select | Shift/Ctrl: add to selection, queue commands
- Right-click: contextual commands (move/attack/gather/heal based on target)
- Control groups 1-9: save and recall unit groups
- Camera: WASD/edge-scroll pan, mouse wheel zoom
- Rally point setting for production buildings

# User Interface

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IMGUI-based UI with dark navy and golden theme. Multiple specialized panels, HUD elements, overlay displays, and full menu system (main menu, skirmish lobby, multiplayer lobby, options).

- Resource HUD: top bar showing Supplies, Iron, Crystal, Veilsteel
- Entity Info Panel: portrait, stats, multi-selection army breakdown
- Action Panel: contextual ability buttons for selected entities
- Build Panel: building placement with cost display and shift-click repeat
- Tech Tree Panel: research tree with costs, prerequisites, queue
- Minimap, floating health bars, income displays, rally markers, movement lines

# Victory Conditions

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Faction elimination - polls building ownership every 2 seconds with a 10-second grace period. A faction is eliminated when it has zero completed buildings. Game ends when one faction survives.

- Only completed buildings count (under-construction excluded)
- 10-second grace period prevents premature elimination during raids
- Disabled in Sandbox and BattalionTest modes
- Post-game stats screen shows match statistics

# Multiplayer Foundation

[IN PROGRESS]

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Multiplayer infrastructure built into core architecture through CommandRouter's lockstep queueing. All commands flow through the same pipeline enabling deterministic synchronization. The networking transport layer is pending implementation.

- CommandRouter supports Remote Player command source
- Lockstep command queueing ensures deterministic game state
- Multiplayer lobby UI complete
- Networking transport layer: pending implementation
- Observer mode infrastructure in GameSettings

# 3. UNIT ROSTER

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35 unique unit types across 6 categories. All stats sourced directly from entity factory code. Units without an Attack Cooldown value have no standard combat component (non-combatants or use custom attack systems).

**Stat Legend:** HP = Hit Points | Dmg = Damage per hit | Spd = Move Speed | LoS = Line of Sight range | CD = Attack Cooldown (seconds between attacks) | Pop = Population cost | Aim = Aim time before firing (ranged) | Range = Min-Max attack range (ranged/magic) | Armor = Armor type category | Dmg Type = Damage type dealt

**Armor Types:** InfantryLight (low protection), InfantryHeavy (good all-round), Cavalry (mounted), Ranged (archers/mages), Structure (siege engines)



# Swordsman

## Core Unit

Primary frontline melee infantry. The backbone of any army with balanced HP and damage. Available to all factions from Era 1, the Swordsman is reliable in all situations but excels at none.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 120           | 12     | 3.5   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.2s          | 1      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryHeavy | 0.5    |       |          |

## Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 1         | 0          | 0         | 0         |

# Archer

## Core Unit

Standard ranged infantry that attacks from distance using pierce projectiles. Automatically retreats when enemies close within minimum range (10). Essential for early defense and supporting melee frontlines.

|          |          |        |        |
|----------|----------|--------|--------|
| HP       | Damage   | Speed  | LoS    |
| 60       | 8        | 4.0    | 25     |
| Cooldown | Aim      | Range  | Pop    |
| 1.5s     | 0.5s     | 10-25  | 1      |
| Class    | Dmg Type | Armor  | Radius |
| Ranged   | Ranged   | Ranged | 0.5    |

## Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 0         | 1          | 0         | 0         |

Tags: ArcherTag

# Builder

## Core Unit

Economy unit that constructs and repairs buildings. Negligible combat stats and no attack cooldown component - not intended for fighting. Auto-chains to nearby unfinished structures within line-of-sight.

|               |        |         |          |
|---------------|--------|---------|----------|
| HP            | Damage | Speed   | LoS      |
| 60            | 2      | 4.0     | 12       |
| Cooldown      | Pop    | Class   | Dmg Type |
| -             | 1      | Economy | Melee    |
| Armor         | Radius |         |          |
| InfantryLight | 0.5    |         |          |

**Special:** No AttackCooldown component. Has CanBuild component for construction.

**Tags:** CanBuild

# Miner

## Core Unit

Resource gatherer that collects Iron and Crystal from deposits. Carries resources back to Hall or Gatherer Hut. Auto-finds new deposits within line-of-sight when depleted. Not a combat unit.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 50            | 2      | 3.5   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| -             | 1      | Miner | Melee    |
| Armor         | Radius |       |          |
| InfantryLight | 0.5    |       |          |

**Special:** No AttackCooldown component. GatherSpeedMultiplier: 1.0 (tech-modifiable). CarryCapacityBonus: 0.

**Tags:** MinerTag

# Scout

## Core Unit

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Fast reconnaissance unit with exceptional vision range (20). Extremely fragile in combat with no attack cooldown component. Invaluable for early exploration and early warning of attacks.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 40            | 3      | 6.0   | 20       |
| Cooldown      | Pop    | Class | Dmg Type |
| -             | 1      | Scout | Melee    |
| Armor         | Radius |       |          |
| InfantryLight | 0.5    |       |          |

**Special:** No AttackCooldown component. Fastest base unit in the game.

# Litharch

## Core Unit

Support healer that restores HP to friendly units. Automatically seeks wounded allies within heal range. Deals magic-type damage in self-defense. Essential for sustaining armies in prolonged engagements.

|          |        |         |          |
|----------|--------|---------|----------|
| HP       | Damage | Speed   | LoS      |
| 60       | 5      | 3.5     | 10       |
| Cooldown | Pop    | Class   | Dmg Type |
| 1.5s     | 1      | Support | Magic    |
| Armor    | Radius |         |          |
| Ranged   | 0.5    |         |          |

## Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 0         | 0          | 0         | 2         |

**Special:** Heal Rate: 8 HP/sec. Heal Range: 4. Has LitharchState for autonomous healing AI.

**Tags:** LitharchTag

# Raider

## Runai Culture

Fast cavalry unit excelling at hit-and-run raids on enemy economies. High speed (7.2) enables rapid map control, caravan interception, and flanking maneuvers. The Runai's primary mobile strike force.

|          |        |       |          |
|----------|--------|-------|----------|
| HP       | Damage | Speed | LoS      |
| 150      | 18     | 7.2   | 10       |
| Cooldown | Pop    | Class | Dmg Type |
| 1.0s     | 1      | Melee | Melee    |
| Armor    | Radius |       |          |
| Cavalry  | 0.6    |       |          |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 1         | 0          | 0         | 0         |

Tags: CavalryTag

# Spearman

## Runai Culture

Anti-cavalry specialist with +50% damage bonus against mounted units. Essential counter-unit in the Runai roster, neutralizing enemy cavalry charges and protecting ranged units from flanking.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 130           | 12     | 5.6   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.0s          | 1      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryHeavy | 0.5    |       |          |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 2         | 1          | 0         | 0         |

**Special:** +50% bonus damage vs Cavalry units (SpearmanTag).

**Tags:** SpearmanTag



# Skirmisher

## Runai Culture

Fast ranged hit-and-run unit. Higher mobility than standard archers enables kiting and harassment. Auto-retreats when enemies close within minimum range (5).

|          |          |        |        |
|----------|----------|--------|--------|
| HP       | Damage   | Speed  | LoS    |
| 95       | 15       | 6.0    | 15     |
| Cooldown | Aim      | Range  | Pop    |
| 1.3s     | 0.5s     | 5-11   | 1      |
| Class    | Dmg Type | Armor  | Radius |
| Ranged   | Ranged   | Ranged | 0.5    |

## Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 0         | 1          | 0         | 0         |

Tags: ArcherTag

# Sand Ballista

## Runai Culture

Runai siege weapon with devastating long-range damage. Slow-firing but deadly during prolonged sieges. Requires melee escort due to vulnerability at close range.

|          |          |           |        |
|----------|----------|-----------|--------|
| HP       | Damage   | Speed     | LoS    |
| 200      | 36       | 3.4       | 22     |
| Cooldown | Aim      | Range     | Pop    |
| 3.5s     | 1.0s     | 8-20      | 2      |
| Class    | Dmg Type | Armor     | Radius |
| Siege    | Siege    | Structure | 0.8    |

## Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 0         | 0          | 2         | 0         |

Tags: ArcherTag, SiegeTag

# Caravan

## Runai Culture

Autonomous trade unit traveling between Trade Hubs and Outposts generating supplies income. Fully autonomous and uncontrollable. Key component of the Runai economic engine.

|          |        |         |               |
|----------|--------|---------|---------------|
| HP       | Damage | Speed   | LoS           |
| 120      | -      | 5.6     | 8             |
| Cooldown | Pop    | Class   | Armor         |
| -        | 0      | Economy | InfantryLight |
| Radius   |        |         |               |
| 0.4      |        |         |               |

**Special:** Uncontrollable. Drops 50% cargo on death. Auto-spawns from Trade Hubs (max 3/route, every 22s). No combat capability.

**Tags:** CaravanTag

# Caravan Escort

## Runai Culture

Auto-assigned bodyguard following a paired Caravan. Engages enemies autonomously. Dies when its Caravan is destroyed. Cannot be commanded independently by the player.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 110           | 10     | 6.2   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.5s          | 0      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryLight | 0.45   |       |          |

**Special:** Uncontrollable. Linked to Caravan - dies when Caravan is destroyed.

**Tags:** CaravanEscortTag

# Sentinel

## Alanthor Culture

Extremely durable heavy infantry with the highest defense values in the game (8/4/2/1). The definitive tank - slow but nearly impossible to bring down. The immovable backbone of Alanthor defensive formations.

| HP                   | Damage     | Speed        | LoS          |
|----------------------|------------|--------------|--------------|
| <b>160</b>           | <b>14</b>  | <b>3.2</b>   | <b>10</b>    |
| Cooldown             | Pop        | Class        | Dmg Type     |
| <b>1.4s</b>          | <b>1</b>   | <b>Melee</b> | <b>Melee</b> |
| Armor                | Radius     |              |              |
| <b>InfantryHeavy</b> | <b>0.5</b> |              |              |

### Defense Values

| Melee Def | Ranged Def | Siege Def | Magic Def |
|-----------|------------|-----------|-----------|
| <b>8</b>  | <b>4</b>   | <b>2</b>  | <b>1</b>  |

**Special:** Highest defense values in the game: 8 Melee / 4 Ranged / 2 Siege / 1 Magic.

# Crossbowman

## Alanthor Culture

Armored ranged unit with exceptional range (22) and heavy damage per shot (16). Slower firing than standard archers but significantly more dangerous and more durable. Excels behind Sentinel lines.

|          |          |               |        |
|----------|----------|---------------|--------|
| HP       | Damage   | Speed         | LoS    |
| 100      | 16       | 3.5           | 25     |
| Cooldown | Aim      | Range         | Pop    |
| 2.0s     | 0.5s     | 6-22          | 1      |
| Class    | Dmg Type | Armor         | Radius |
| Ranged   | Ranged   | InfantryHeavy | 0.5    |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 2         | 2          | 0         | 0         |

Tags: ArcherTag

# Cataphract

## Alanthor Culture

Elite heavy cavalry combining high damage (18) with substantial armor. Expensive at 2 pop but devastating on the charge. Merges Alanthor's defensive resilience with cavalry mobility for powerful flanking.

|          |        |       |          |
|----------|--------|-------|----------|
| HP       | Damage | Speed | LoS      |
| 180      | 18     | 6.5   | 10       |
| Cooldown | Pop    | Class | Dmg Type |
| 1.1s     | 2      | Melee | Melee    |
| Armor    | Radius |       |          |
| Cavalry  | 0.6    |       |          |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 3         | 2          | 0         | 0         |

Tags: CavalryTag

# Ballista

## Alanthor Culture

The longest-range siege weapon in the game (range 24, LoS 26) with the highest single-shot damage (40). Extremely slow to reload (4.0s) but devastating against structures and clustered units.

|          |          |           |        |
|----------|----------|-----------|--------|
| HP       | Damage   | Speed     | LoS    |
| 220      | 40       | 2.8       | 26     |
| Cooldown | Aim      | Range     | Pop    |
| 4.0s     | 1.0s     | 10-24     | 2      |
| Class    | Dmg Type | Armor     | Radius |
| Siege    | Siege    | Structure | 0.8    |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 0         | 0          | 3         | 0         |

Tags: ArcherTag, SiegeTag



# Warboar Rider

## Feraldis Culture

Fast aggressive mounted unit on tamed warboars. Cheaper than other cavalry (1 pop), enabling swarm tactics. Complements Feraldis's aggressive, numbers-driven playstyle.

|          |        |       |          |
|----------|--------|-------|----------|
| HP       | Damage | Speed | LoS      |
| 160      | 16     | 7.0   | 10       |
| Cooldown | Pop    | Class | Dmg Type |
| 1.0s     | 1      | Melee | Melee    |
| Armor    | Radius |       |          |
| Cavalry  | 0.6    |       |          |

## Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 1         | 0          | 0         | 0         |

Tags: CavalryTag

# Hunter

## Feraldis Culture

Mobile ranged unit specializing in hit-and-run warfare. Faster than standard archers (5.7) with shorter range (4-12), rewarding aggressive positioning and flanking maneuvers.

|          |          |        |        |
|----------|----------|--------|--------|
| HP       | Damage   | Speed  | LoS    |
| 100      | 11       | 5.7    | 15     |
| Cooldown | Aim      | Range  | Pop    |
| 1.2s     | 0.5s     | 4-12   | 1      |
| Class    | Dmg Type | Armor  | Radius |
| Ranged   | Ranged   | Ranged | 0.5    |

Tags: ArcherTag

# Siege Ram

## Feraldis Culture

Massive melee siege engine with the highest HP of any non-Crystal unit (300). Purpose-built for smashing buildings. Very slow (3.0) but incredibly durable with heavy siege-resistant armor.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 300           | 34     | 3.0   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 3.0s          | 2      | Siege | Siege    |
| Armor         | Radius |       |          |
| InfantryHeavy | 1.0    |       |          |

## Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 4         | 2          | 6         | 0         |

Tags: SiegeTag

# Berserker

## Feraldis Culture

Converted from Miners at the Fiendstone Keep. Trades healing and heavy armor for raw attack power (14) and speed (5.8). Cannot be healed. A desperate but powerful glass cannon gamble.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 80            | 14     | 5.8   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.0s          | 1      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryLight | 0.5    |       |          |

## Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 2         | 0          | 0         | 0         |

**Special:** Unhealable - cannot be healed by Litharchs. Converted from Miners, not trained.

**Tags:** BerserkerTag, UnhealableTag

# Ashblade

## Sect Unit - Ember Ash

Light melee infantry with above-average speed (5.0) for rapid strikes and hit-and-run. Trades durability for mobility and attack frequency (1.0s cooldown).

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 155           | 14     | 5.0   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.0s          | 1      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryLight | 0.5    |       |          |

Tags: SectUniqueUnitTag

# Brandbreaker

## Sect Unit - Hollow Brand

Anti-structure specialist dealing siege-type damage in melee, making it effective against buildings without being a slow traditional siege engine. Has light armor for survivability.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 150           | 12     | 4.0   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.5s          | 1      | Siege | Siege    |
| Armor         | Radius |       |          |
| InfantryHeavy | 0.5    |       |          |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 1         | 1          | 0         | 0         |

**Special:** Deals Siege damage in melee - highly effective vs buildings.

**Tags:** SiegeTag, SectUniqueUnitTag

# Flame Warden

## Sect Unit - Still Flame

Balanced melee fighter with well-rounded stats across HP (150), damage (15), and speed (3.8). A reliable frontline unit that fits any army composition without weakness.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 150           | 15     | 3.8   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.1s          | 1      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryHeavy | 0.5    |       |          |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 1         | 1          | 0         | 0         |

Tags: SectUniqueUnitTag

# Chaincaster

## Sect Unit - Flamewrought Chains

Control-oriented magic caster dealing magic damage at moderate range (14), bypassing physical armor entirely. Lower raw damage (10) compensated by magic type's armor-piercing effectiveness.

| HP           | Damage       | Speed         | LoS        |
|--------------|--------------|---------------|------------|
| <b>105</b>   | <b>10</b>    | <b>3.5</b>    | <b>18</b>  |
| Cooldown     | Aim          | Range         | Pop        |
| <b>1.8s</b>  | <b>0.5s</b>  | <b>0-14</b>   | <b>1</b>   |
| Class        | Dmg Type     | Armor         | Radius     |
| <b>Magic</b> | <b>Magic</b> | <b>Ranged</b> | <b>0.5</b> |

## Defense Values

| Melee Def | Ranged Def | Siege Def | Magic Def |
|-----------|------------|-----------|-----------|
| <b>0</b>  | <b>0</b>   | <b>0</b>  | <b>1</b>  |

**Tags:** SectUniqueUnitTag, ArcherTag



# Archivist Adept

## Sect Unit - Veiled Memory

Scholarly mage with fast casting (0.4s aim) and strong damage output (14). Good balance between survivability (110 HP) and offensive magic, with moderate magic resistance.

|          |          |        |        |
|----------|----------|--------|--------|
| HP       | Damage   | Speed  | LoS    |
| 110      | 14       | 3.5    | 18     |
| Cooldown | Aim      | Range  | Pop    |
| 1.6s     | 0.4s     | 0-14   | 1      |
| Class    | Dmg Type | Armor  | Radius |
| Magic    | Magic    | Ranged | 0.5    |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 0         | 0          | 0         | 2         |

Tags: SectUniqueUnitTag, ArcherTag

# Glassmark Arcanist

## Sect Unit - Mirror Rite

Glass cannon mage with the highest magic damage (18) and longest range (15) among sect units, but the most fragile (100 HP). Devastating when protected behind melee lines.

|          |          |        |        |
|----------|----------|--------|--------|
| HP       | Damage   | Speed  | LoS    |
| 100      | 18       | 3.5    | 19     |
| Cooldown | Aim      | Range  | Pop    |
| 1.5s     | 0.4s     | 0-15   | 1      |
| Class    | Dmg Type | Armor  | Radius |
| Magic    | Magic    | Ranged | 0.5    |

## Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 0         | 0          | 0         | 1         |

Tags: SectUniqueUnitTag, ArcherTag

# Golem Autark

## Sect Unit - Antiquity

Massive magical construct - the tankiest sect unit at 320 HP with heavy defense (3/2/1/2). Extremely slow (2.0) but deals high magic damage (22) at range. A walking fortress.

|          |          |               |        |
|----------|----------|---------------|--------|
| HP       | Damage   | Speed         | LoS    |
| 320      | 22       | 2.0           | 14     |
| Cooldown | Aim      | Range         | Pop    |
| 2.0s     | 0.6s     | 0-10          | 1      |
| Class    | Dmg Type | Armor         | Radius |
| Magic    | Magic    | InfantryHeavy | 0.6    |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 3         | 2          | 1         | 2         |

Tags: SectUniqueUnitTag, ArcherTag

# Judicator

## Sect Unit - Shard Judgment

Heavy melee enforcer with strong offensive (16 damage) and defensive stats (2/1 defense). Effective at holding key positions and anchoring defensive formations.

|                      |            |              |              |
|----------------------|------------|--------------|--------------|
| HP                   | Damage     | Speed        | LoS          |
| <b>160</b>           | <b>16</b>  | <b>3.4</b>   | <b>10</b>    |
| Cooldown             | Pop        | Class        | Dmg Type     |
| <b>1.2s</b>          | <b>1</b>   | <b>Melee</b> | <b>Melee</b> |
| Armor                | Radius     |              |              |
| <b>InfantryHeavy</b> | <b>0.5</b> |              |              |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| <b>2</b>  | <b>1</b>   | <b>0</b>  | <b>0</b>  |

Tags: SectUniqueUnitTag

# Nullblade

## Sect Unit - Unmaker's Grasp

Anti-Crystal specialist with higher speed (4.2) to chase Crystal creatures. Light armor with magic resistance (1). Purpose-designed to counter the Crystal faction's encroaching threat.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 150           | 14     | 4.2   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.1s          | 1      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryLight | 0.5    |       |          |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 0         | 0          | 0         | 1         |

Tags: SectUniqueUnitTag

# Scar Guard

## Sect Unit - Renewal

Tough frontline fighter with 170 HP - the highest among standard sect melee units. Good damage (16) with solid defense (2/1). An excellent damage sponge and formation anchor.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 170           | 16     | 3.2   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.2s          | 1      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryHeavy | 0.5    |       |          |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 2         | 1          | 0         | 0         |

Tags: SectUniqueUnitTag

# Stone Warden

## Sect Unit - Living Stone

Supreme tank at 200 HP with heavy defense (3/2/1/0) but lowest damage (10) and speed (2.8) of any sect unit. Built purely to absorb punishment while allies deal damage.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 200           | 10     | 2.8   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.4s          | 1      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryHeavy | 0.5    |       |          |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 3         | 2          | 1         | 0         |

Tags: SectUniqueUnitTag

# Vault Keeper

## Sect Unit - Quiet Vault

Defensive guardian with moderate stats across the board (140 HP, 12 damage, 3.5 speed). Solid defense (2/1) makes it a reliable but unspectacular melee fighter for protecting assets.

|               |        |       |          |
|---------------|--------|-------|----------|
| HP            | Damage | Speed | LoS      |
| 140           | 12     | 3.5   | 10       |
| Cooldown      | Pop    | Class | Dmg Type |
| 1.3s          | 1      | Melee | Melee    |
| Armor         | Radius |       |          |
| InfantryHeavy | 0.5    |       |          |

### Defense Values

|           |            |           |           |
|-----------|------------|-----------|-----------|
| Melee Def | Ranged Def | Siege Def | Magic Def |
| 2         | 1          | 0         | 0         |

Tags: SectUniqueUnitTag



# Crystalling

## Crystal Faction

Swarming melee creature spawned from Crystal Nodes. Individually weak (60 HP, 8 damage) but fastest attack speed in the game (0.8s) and produced in overwhelming numbers at low cost (35 crystal).

|          |               |              |       |
|----------|---------------|--------------|-------|
| HP       | Damage        | Speed        | LoS   |
| 60       | 8             | 5.5          | 10    |
| Cooldown | Pop           | Crystal Cost | Class |
| 0.8s     | 0             | 35           | Melee |
| Dmg Type | Armor         | Radius       |       |
| Melee    | InfantryLight | 0.4          |       |

**Special:** No population cost - uses crystal resource economy. Spawned by CrystalAISystem.

**Tags:** CrystalTag, CrystalUnitTag

# Veilstinger

## Crystal Faction

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Crystal ranged attacker with dual-laser targeting, striking 2 enemies simultaneously. Extreme range (8-24), highest LoS in game (28), and high damage (18) but paper-thin (40 HP). Uses custom VeilstingerState instead of standard ArcherState.

| HP           | Damage        | Speed        | LoS           |
|--------------|---------------|--------------|---------------|
| <b>40</b>    | <b>18</b>     | <b>4.0</b>   | <b>28</b>     |
| Cooldown     | Aim           | Range        | Pop           |
| <b>-</b>     | <b>0.4s</b>   | <b>8-24</b>  | <b>0</b>      |
| Crystal Cost | Class         | Dmg Type     | Armor         |
| <b>80</b>    | <b>Ranged</b> | <b>Magic</b> | <b>Ranged</b> |

**Special:** Dual-target laser attack (VeilstingerState). Hits 2 enemies simultaneously. Uses custom attack system (no standard AttackCooldown). Damage type is Magic.

**Tags:** CrystalTag, CrystalUnitTag

# Godsplinter

## Crystal Faction

Colossal crystal siege monster - the most powerful unit in the game. 1200 HP with massive defense (10/8/5/5). Dual attack modes: melee siege (range 4, 2x vs buildings) and laser barrage (range 22, 4 targets). Uses custom GodsplinterState.

| HP           | Damage      | Speed         | LoS           |
|--------------|-------------|---------------|---------------|
| 1200         | 40          | 1.8           | 20            |
| Siege Range  | Laser Range | Laser Targets | Pop           |
| 4            | 22          | 4             | 0             |
| Crystal Cost | Class       | Dmg Type      | Armor         |
| 350          | Siege       | Siege         | InfantryHeavy |

### Defense Values

| Melee Def | Ranged Def | Siege Def | Magic Def |
|-----------|------------|-----------|-----------|
| 10        | 8          | 5         | 5         |

**Special:** Custom GodsplinterState (no standard AttackCooldown). Melee: 2x damage vs buildings. Laser barrage: hits up to 4 targets at range 22. Highest defense values of any unit: 10/8/5/5.

**Tags:** CrystalTag, CrystalUnitTag